MSDI Calculator

Crop and Soil Coefficients

Methodology

Technical Methodology Memorandum

Prepared for internal model documentation and technical review

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Model scope: Manure Sub-Surface Drip Irrigation (mSDI-e) economic screening calculator

Document status: Methodology and documentation of crop and soil coefficients used within the MSDI calculator.

Purpose and Scope

This memorandum documents the crop and soil coefficient tables used by the MSDI Calculator to support crop water demand, irrigation requirement, nutrient demand, salinity screening, and soil-water calculations. The document is coefficient-focused: it explains the values, screening-level basis, revision logic, and implementation implications for crop and soil lookup tables. It is intended to complement the separate Crop Water Demand and Irrigation Requirement Methodology, which documents the calculation sequence that uses these coefficients.

The coefficient tables are intended for planning-level comparison of current systems and manure subsurface drip irrigation (mSDI-e), not for field-specific irrigation scheduling, nutrient management planning, engineering design, or regulatory reporting.

Source Lookup Tables and Fields

Lookup table	Relevant fields	Primary use in model
Crops_LU	Crop type; seasonal growing days; seasonal Kc; TN, P2O5, and K2O seasonal demand	Converts seasonal ET0 into crop water demand and supports crop nutrient requirement calculations.
Soils_LU	Soil texture class; salinity risk; CEC; leaching fractions; allowable ECw/TDS; effective precipitation coefficients; emitter spacing	Supports effective rainfall, salt/leaching screening, irrigation design assumptions, and future environmental-impact calculations.

Terminology and Units

Term	Definition in this methodology	Typical unit
ET0	Seasonal reference evapotranspiration supplied by the Weather_LU table.	inches per season
Kc	Crop coefficient used to scale ET0 to crop evapotranspiration.	dimensionless
ETc	Crop water demand or crop evapotranspiration calculated as ET0 x Kc.	inches per season
Growing days	Number of days in a season assigned to a crop for seasonal weighting.	days per season
Effective precipitation coefficient	Fraction of seasonal precipitation assumed to contribute to crop water supply.	dimensionless
Nutrient demand	Seasonal crop demand for N, P2O5, and K2O.	lb/ac/season

Part 1. Crop Coefficient Methodology

The crop coefficient approach is an established irrigation-planning method. FAO-56 separates climate demand from crop response: reference evapotranspiration represents the evaporative demand of the atmosphere, while Kc represents crop-specific canopy, aerodynamic, and soil evaporation effects. The MSDI Calculator uses this screening-level Kc x ET0 structure rather than a daily dual-coefficient soil-water model.

Equation 1. $ETc_{s,c} = ET0_s \times Kc_{s,c}$

where $ETc_{s,c}$ is seasonal crop water demand for season s and crop c, $ET0_s$ is seasonal reference evapotranspiration, and $Kc_{s,c}$ is the seasonal crop coefficient.

1.1 Seasonal Crop Weighting

The lookup table allows a crop to occupy part or all of a season. If a crop does not occupy the full season, the remaining portion is assigned to the next seasonal crop in the rotation. This allows the model to represent crop transitions without requiring the user to enter planting and harvest dates.

Equation 2. ETc_s = (GD_s,c / D_s) x (Kc_s,c x ET0_s) + ((D_s - GD_s,c) / D_s) x (Kc_s,next x ET0_s)

where GD_s,c is the growing days assigned to the current crop in season s, D_s is the standardized season length, Kc_s,c is the current crop coefficient for that season, Kc_s,next is the next crop coefficient used for the remainder of the season, and ET0_s is seasonal reference evapotranspiration.

1.2 Implementation-Specific Seasonal Equations

Spring: ETc_spring = (GD_spring,springCrop / D) x (Kc_spring,springCrop x ET0_spring) + ((D - GD_spring,springCrop) / D) x (Kc_spring,summerCrop x ET0_spring)

Summer: ETc_summer = (GD_summer,summerCrop / D) x (Kc_summer,summerCrop x ET0_summer) + ((D - GD_summer,summerCrop) / D) x (Kc_summer,fallCrop x ET0_summer)

Fall: ETc_fall = (GD_fall,fallCrop / D) x (Kc_fall,fallCrop x ET0_fall) + ((D - GD_fall,fallCrop) / D) x (Kc_fall,winterCrop x ET0_fall)

Winter: ETc_winter = (GD_winter,winterCrop / D) x (Kc_winter,winterCrop x ET0_winter) + ((D - GD_winter,winterCrop) / D) x (Kc_winter,springCrop x ET0_winter)

1.3 Crop Coefficient and Growing-Day Lookup Values

Table 1 summarizes the current crop growing-day and seasonal Kc values. These values are screening-level coefficients and should not be interpreted as daily crop-stage curves or site-specific irrigation recommendations.

Table 1. Current crop growing-day and seasonal crop coefficient values in Crops_LU.

Crop	Fall days	Winter days	Spring days	Summer days	Fall Kc	Winter Kc	Spring Kc	Summer Kc
N/A (Fallow)	0	0	0	0	0	0	0	0
Alfalfa	91.25	91.25	91.25	91.25	0.9	0.65	0.85	1.15
Barley	91.25	0	91.25	0	0.7	0	0.9	0
Brassica	40	0	60	91.25	0.6	0	0.7	1.05
Canola	40	0	60	91.25	0.6	0	0.7	1.05
Clover	40	0	60	91.25	0.6	0	0.7	1.05
Corn	20	0	60	91.25	0.5	0	0.7	1.2
Cottonseed	40	0	60	91.25	0.7	0	0.6	1.15
Fescue	91.25	91.25	91.25	91.25	0.9	0.65	0.85	1.15
Field peas / beans	40	0	60	91.25	0.6	0	0.7	1.05
Millet	40	0	60	91.25	0.6	0	0.7	1.05
Mixed grass-legume Pasture	91.25	91.25	91.25	91.25	0.9	0.65	0.85	1.15
Oats	40	0	60	91.25	0.6	0	0.7	1.05
Orchardgrass	91.25	91.25	91.25	91.25	0.9	0.65	0.85	1.15
Perennial ryegrass	40	0	60	91.25	0.6	0	0.7	1.05
Sorghum	40	0	60	91.25	0.7	0	0.6	1
Soybeans	30	0	50	91.25	0.6	0	0.7	1.05
Sudangrass	91.25	91.25	91.25	91.25	0.9	0.65	0.85	1.15

Sugar beets / Fodder beets	40	0	60	91.25	0.6	0	0.7	1.05
Sunflower	40	0	60	91.25	0.7	0	0.6	1
Timothy	40	0	60	91.25	0.6	0	0.7	1.05
Triticale	40	0	60	91.25	0.6	0	0.7	1.05
Wheat	0	91.25	91.25	0	0	0.75	0.9	0

1.4 Crop Nutrient Demand Coefficients

Crops_LU also stores seasonal nutrient demand values for total nitrogen, phosphorus as P2O5, and potassium as K2O. These are used by nutrient-balancing portions of the application and should be interpreted as screening-level seasonal demand values, not site-specific nutrient management recommendations.

Table 2. Current crop nutrient demand coefficients in Crops_LU.

Crop	TN lb/ac/season	P2O5 lb/ac/season	K2O lb/ac/season
N/A (Fallow)	0	0	0
Alfalfa	0	80	200
Barley	80	35	50
Brassica	90	60	120
Canola	140	60	80
Clover	0	60	150
Corn	160	70	80
Cottonseed	90	60	90
Fescue	100	50	100
Field peas / beans	0	40	40
Millet	60	30	40
Mixed grass-legume Pasture	50	40	80
Oats	60	30	40
Orchardgrass	120	50	120
Perennial ryegrass	150	50	150
Sorghum	100	40	80
Soybeans	0	60	80
Sudangrass	180	60	120
Sugar beets / Fodder beets	120	80	120
Sunflower	95	40	80
Timothy	100	40	100
Triticale	100	40	60
Wheat	100	40	60

Part 2. Soil Coefficient Methodology

Soil texture affects water storage, infiltration, runoff potential, drainage, salt accumulation, and the fraction of rainfall that can be stored in the root zone for crop use. NRCS describes available water capacity as the quantity of water a soil can store for plant use, with major controls including texture, organic matter, bulk density, and structure. For a screening model, soil texture class is an appropriate first-order basis for assigning soil-water coefficients, while recognizing that field-specific design should use site-specific soil survey data where available.

2.1 Effective Precipitation Coefficients

Effective precipitation is the fraction of seasonal precipitation assumed to reduce irrigation requirement.

Equation 3. Peff_s = P_s x f_eff,soil,s

where Peff_s is effective precipitation in season s, P_s is seasonal precipitation from Weather_LU, and f_eff,soil,s is the soil- and season-specific effective precipitation coefficient from Soils_LU.

The original lookup table included separate P_eff_Fall, P_eff_Winter, P_eff_Spring, and P_eff_Summer columns. However, the uploaded implementation files for effective rainfall use the P_EFF_WINTER lookup column for all four seasons. The app should be updated to use the season-specific fields if the revised coefficients are implemented.

2.2 Review and Revision of Effective Precipitation Coefficients

The existing values generally increased effective precipitation for coarser soils and decreased it for finer soils. This was directionally reasonable for runoff risk on fine-textured soils, but it underrepresented the lower storage capacity and greater deep-percolation potential of very coarse soils. A more defensible screening structure is bell-shaped across texture classes: loam and silt loam generally receive the highest effective fraction, sands are reduced for storage limitations, and heavy clays are reduced for runoff, slow infiltration, and waterlogging losses.

The revised values were generated using a transparent two-factor screening equation:

Equation 4. f_eff,soil,s = round(B_s x T_soil, 0.01)

where B_s is a seasonal base coefficient and T_soil is a soil texture multiplier. The seasonal base coefficient reflects the relative likelihood that rainfall is beneficial during the season, while the soil multiplier reflects texture-driven storage, infiltration, and runoff behavior.

Table 3. Seasonal base coefficients used for revised effective precipitation values.

Season	Base coefficient	Rationale
Fall	0.65	Moderate crop demand and moderate probability that rainfall contributes to root-zone storage.
Winter	0.45	Lowest effective fraction because crop demand is often lower and soils may already be wet, increasing runoff or deep percolation losses.
Spring	0.67	Moderate-high effective fraction due active crop growth and increasing evaporative demand.
Summer	0.75	Highest effective fraction because crop demand and evaporative demand are highest, increasing the probability that precipitation offsets irrigation requirement.

Table 4. Soil texture multipliers used for revised effective precipitation values.

Soil texture	Texture multiplier	Rationale
Sand	0.85	Reduced by low water-holding capacity and rapid deep percolation.
Loamy sand	0.9	Slightly higher storage than sand but still coarse-textured.
Sandy loam	0.96	Moderate storage and infiltration; near baseline.
Loam	1.04	Highest effective fraction due favorable balance of infiltration and storage.
Silt loam	1.0	High storage, but potential crusting/runoff offsets part of the benefit.
Silt	0.97	Good storage but higher runoff/crusting sensitivity than loam.
Sandy clay loam	0.94	Moderate storage with some fine-texture runoff limitation.

Clay loam	0.89	Reduced for slower infiltration and runoff/waterlogging risk.
Silty clay loam	0.86	Reduced for fine texture and slower infiltration.
Sandy clay	0.83	Fine-textured behavior with some sand; moderate runoff/storage limitation.
Silty clay	0.78	Reduced for slow infiltration and runoff potential.
Clay	0.74	Lowest effective fraction among fine textures due runoff, slow infiltration, and waterlogging risk.

Table 5. Original and revised soil effective precipitation coefficients.

Soil	Orig Fall	Orig Winter	Orig Spring	Orig Summer	Rev Fall	Rev Winter	Rev Spring	Rev Summer
Sand	0.7	0.45	0.7	0.8	0.55	0.38	0.57	0.64
Loamy sand	0.7	0.45	0.7	0.8	0.59	0.41	0.6	0.68
Sandy loam	0.7	0.45	0.7	0.8	0.62	0.43	0.64	0.72
Loam	0.65	0.4	0.65	0.75	0.68	0.47	0.7	0.78
Silt loam	0.65	0.4	0.65	0.75	0.65	0.45	0.67	0.75
Silt	0.65	0.4	0.65	0.75	0.63	0.44	0.65	0.73
Sandy clay loam	0.65	0.4	0.65	0.75	0.61	0.42	0.63	0.7
Clay loam	0.55	0.3	0.6	0.7	0.58	0.4	0.6	0.67
Silty clay loam	0.55	0.3	0.6	0.7	0.56	0.39	0.58	0.65
Sandy clay	0.55	0.3	0.6	0.7	0.54	0.37	0.56	0.62
Silty clay	0.55	0.3	0.6	0.7	0.51	0.35	0.52	0.58
Clay	0.55	0.3	0.6	0.7	0.48	0.33	0.5	0.55

The revised values remain screening-level coefficients. They should not be interpreted as calibrated hydrologic parameters. Future versions should ideally incorporate regional rainfall-effectiveness assumptions because rainfall effectiveness differs substantially among Mediterranean, humid subtropical, arid, and high-rainfall dairy regions.

Other Soil Coefficients

Soils_LU also includes salt-accumulation risk, CEC, salt retention multipliers, conventional and SDI leaching fractions, allowable ECw and TDS thresholds, break-even salt loading values, SDI leaching factors, and recommended emitter spacing. These support salinity screening, manure-water suitability, and future environmental-impact calculations. They should be documented in greater detail in salinity and environmental-impact methodology chapters.

Field group	Model role	Current status
Salt-accumulation risk and CEC	Qualitative and quantitative indicators of soil buffering and salinity sensitivity.	Retained as screening-level soil texture assumptions.
Leaching fractions	Estimate fraction of water moving below the root zone under conventional and SDI conditions.	Used for salinity and future nitrate-leaching logic; should be reviewed with environmental module.
Allowable ECw/TDS and break-even salt loading	Support screening for salt loading from manure liquid.	Useful but should be documented in a dedicated salinity methodology.
Emitter spacing	Provides a soil-texture-based design assumption for mSDI-e lateral/emitter behavior.	Screening-level design input; not a substitute for engineering design.

Assumptions and Limitations

- Crop coefficients are seasonal screening values and do not represent daily growth-stage Kc curves.
- Growing-day values are simplified seasonal allocation assumptions and do not represent local planting, cutting, dormancy, or harvest calendars.
- Effective precipitation coefficients are dimensionless screening factors and do not simulate storm intensity, antecedent moisture, runoff, drainage, or daily soil-water storage.
- Soil texture is treated as the dominant soil-water descriptor, but actual available water capacity depends on organic matter, bulk density, structure, salinity, compaction, and rock fragments.
- Revised effective precipitation coefficients are intended to improve internal consistency, not to replace regional hydrologic calibration.
- The current implementation should be reviewed to ensure seasonal effective precipitation columns are used correctly.

Processing Workflow Summary

1. Select crop rotation inputs from user responses.
2. Retrieve crop growing days and seasonal Kc values from Crops_LU.
3. Retrieve seasonal ET0 and precipitation from Weather_LU.
4. Calculate seasonal ETc using the weighted crop coefficient equations.
5. Retrieve soil texture from the user-selected soil class.
6. Retrieve season-specific effective precipitation coefficients from Soils_LU.
7. Calculate effective precipitation for each season.
8. Pass ETc and effective precipitation to the crop water demand and irrigation requirement calculations.

References and Source Links

[1] FAO Irrigation and Drainage Paper 56. Crop evapotranspiration: Guidelines for computing crop water requirements. https://www.fao.org/4/X0490E/x0490e00.htm

[2] FAO-56 Chapter 5. Introduction to crop evapotranspiration and the crop coefficient approach. https://www.fao.org/4/X0490E/x0490e0a.htm

[3] USDA NRCS. Part 652 Irrigation Guide. https://www.nrcs.usda.gov/sites/default/files/2023-01/7385.pdf

[4] USDA NRCS. Available Water Storage / Available Water Capacity technical reference. https://www.nrcs.usda.gov/sites/default/files/2023-05/AvailableWaterStorage_0-150.pdf

[5] USDA NRCS. Soil texture calculator and SPAW soil-water characteristics tools. https://www.nrcs.usda.gov/conservation-basics/soil/calculator-tools

[6] USGS. Soil Properties Dataset in the United States. https://www.usgs.gov/data/soil-properties-dataset-united-states